

## 14

Singular Value  
Decomposition

Rotation → Scaling → Rotation Again

Singular value decomposition (SVD) is a universal and powerful tool in the linear algebra toolbox. The eigenvalue decomposition, Cholesky decomposition, and LDL decomposition introduced earlier in this book all impose requirements on the shape and properties of the matrix; SVD, however, imposes no such requirements at all – a real matrix of any shape can undergo singular value decomposition. In this respect it resembles QR decomposition. Unlike QR decomposition, however, SVD reveals the intrinsic structure of the matrix. This broad applicability lets SVD shine in areas such as data compression and feature extraction, among others.

This book laid the groundwork for singular value decomposition when discussing the spectral decomposition of Gram matrices; in this chapter, let's explore singular value decomposition comprehensively.

## 14.1 Singular Value Decomposition



The core skills you'll master in this section:

- ▶ A real matrix of any shape can undergo singular value decomposition.
- ▶ In the full SVD,  $U$  and  $V$  are both orthogonal matrices
- ▶  $U$  and  $V$  represent an orthonormal basis for their respective spaces, and can be viewed as rotation operations.
- ▶  $S$  is a diagonal matrix, whose main diagonal holds the singular values.
- ▶ The rank of the matrix equals the number of nonzero singular values.
- ▶ Solving for the eigenvalues and eigenvectors of two Gram matrices lets us reconstruct the SVD process.

## Full SVD

Given an arbitrary real matrix  $X_{n \times D}$ , performing (full) singular value decomposition (SVD) yields

$$X_{n \times D} = U_{n \times n} S_{n \times D} V_{D \times D}^T$$

where

$U$  is the left-singular vector matrix;  $U$ 's column vectors are the left-singular vectors.  $U$  is an orthogonal matrix,  $U = [u_1, u_2, \dots, u_n]$ , which span an orthonormal basis for the  $\mathbb{R}^n$  space.

$S$  is a diagonal matrix, and its main-diagonal elements are the singular values. Geometrically, singular values represent the scaling effect along different dimensions.

**⚠** Note that a diagonal matrix's off-diagonal elements are all 0; also note that a diagonal matrix need not be square.

$V$  is the right-singular vector matrix;  $V$ 's column vectors are the right-singular vectors.  $V$  is also an orthogonal matrix,  $V = [v_1, v_2, \dots, v_D]$ , which span an orthonormal basis for the  $\mathbb{R}^D$  space.

Figure 1 shows a heatmap of the full SVD decomposition. Figure 2 shows four common types of SVD, which we'll cover later in this chapter.

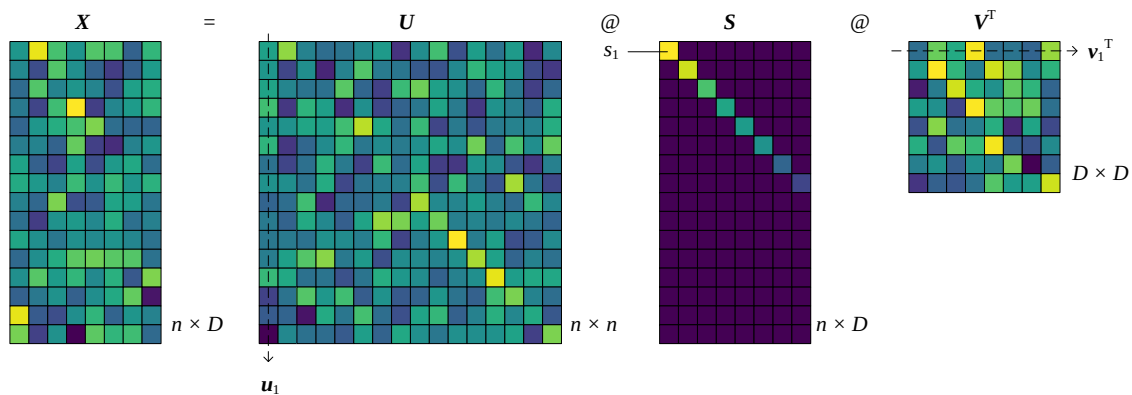


Figure 1. Full SVD

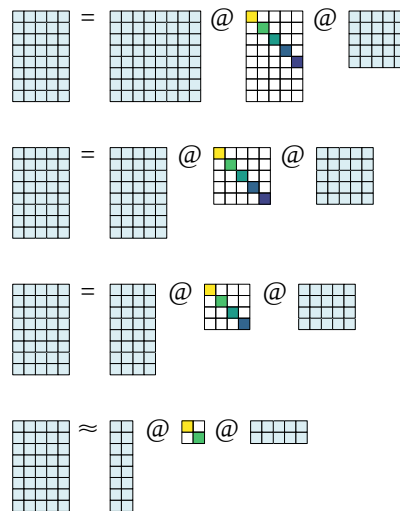


Figure 2. Four Types of SVD

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*S: main-diagonal elements are the singular values*

$S$ 's main-diagonal elements are the singular values, generally arranged in descending order

$$\mathbf{S}_{n \times D} = \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_D \end{bmatrix}$$

If  $\mathbf{X}$  is full rank,  $\text{rank}(\mathbf{X}) = D$ ,  $\mathbf{S}$ 's main-diagonal elements (the singular values  $s_j$ ) are all greater than 0, and their relative sizes are generally:

$$s_1 \geq s_2 \geq \dots \geq s_D > 0$$

This tells us that  $\mathbf{S}$  can be split into two blocks

$$\mathbf{S}_{n \times D} = \begin{bmatrix} \mathbf{S}_{D \times D} \\ \mathbf{O}_{(n-D) \times D} \end{bmatrix}$$

If  $\mathbf{X}$  is not full column rank — say  $\text{rank}(\mathbf{X}) = r$  — then  $\mathbf{X}$  has  $D - r$  singular values equal to 0. In other words, the rank of the matrix equals the number of nonzero singular values.

For example, as shown in Figure 3, if we replace  $\mathbf{X}$ 's last two columns with  $\mathbf{X}$ 's first two columns, then  $\mathbf{X}$  is not full column rank. We find that two of its singular values are 0.

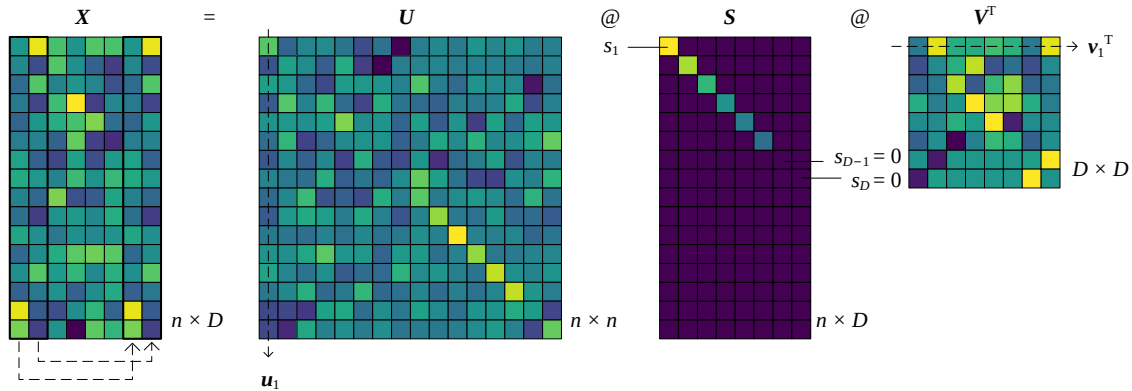


Figure 3. Full SVD,  $\mathbf{X}$  is Not Full Column Rank

### $\mathbf{U}$ : Left-singular vector matrix

In the full SVD, the left-singular vector matrix  $\mathbf{U}$  is an orthogonal matrix, that is, it satisfies

$$\mathbf{U}_{n \times n}^T @ \mathbf{U}_{n \times n} = \mathbf{U}_{n \times n} @ \mathbf{U}_{n \times n}^T = \mathbf{I}_{n \times n}$$

Expand using the first perspective of matrix multiplication:  $\mathbf{U}^T \mathbf{U}$

$$\mathbf{U}_{n \times n}^T @ \mathbf{U}_{n \times n} = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_n \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & \mathbf{u}_1^T \mathbf{u}_2 & \dots & \mathbf{u}_1^T \mathbf{u}_n \\ \mathbf{u}_2^T \mathbf{u}_1 & \mathbf{u}_2^T \mathbf{u}_2 & \dots & \mathbf{u}_2^T \mathbf{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_n^T \mathbf{u}_1 & \mathbf{u}_n^T \mathbf{u}_2 & \dots & \mathbf{u}_n^T \mathbf{u}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

As shown in Figure 4,  $\mathbf{U}$ 's column vectors are unit vectors and pairwise orthogonal. Also, as shown in Figure 5,  $\mathbf{U}$ 's row vectors are likewise all unit vectors and pairwise orthogonal.

In other words, geometrically, the orthogonal matrix  $U$  can be viewed as a rotation in the  $\mathbb{R}^n$  space.

⚠ Note that earlier in this book we mentioned that the identity matrix, rotation matrices, permutation matrices, reflection matrices, and their composite transforms are all orthogonal matrices; for convenience of discussion, we treat the orthogonal matrices in SVD as representing spatial rotations.

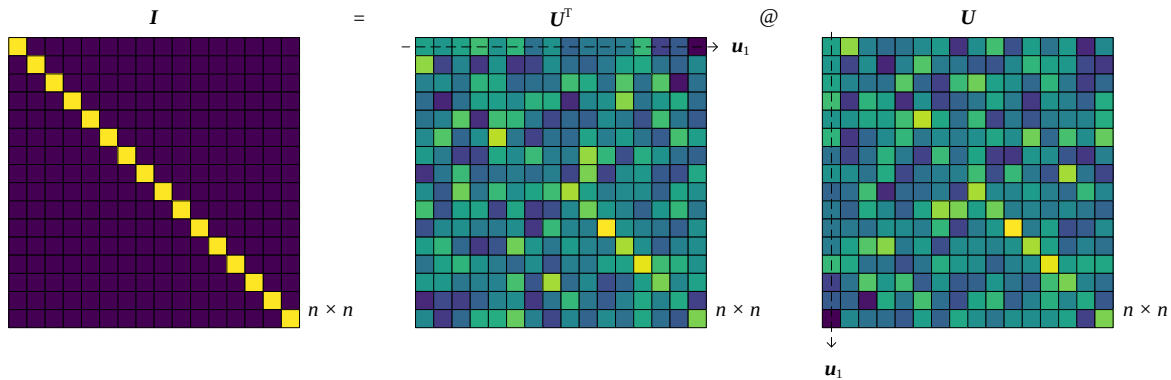


Figure 4.  $U$ 's column vectors are all unit vectors and pairwise orthogonal

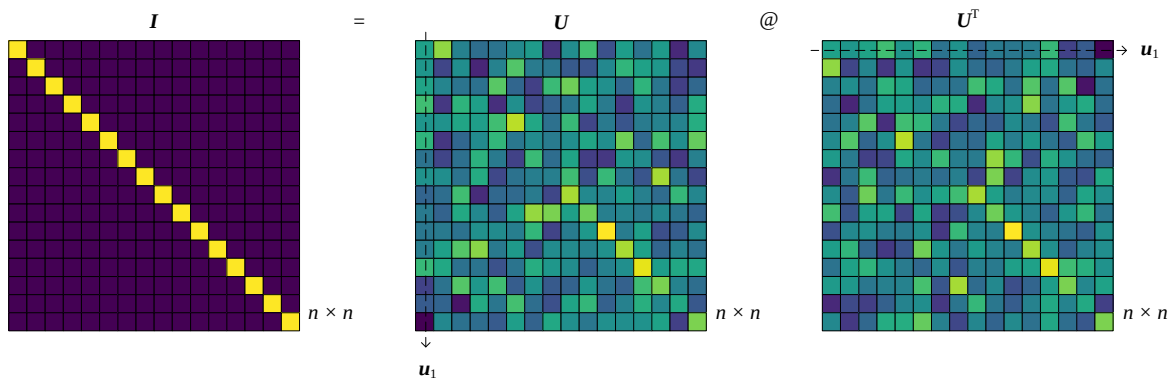


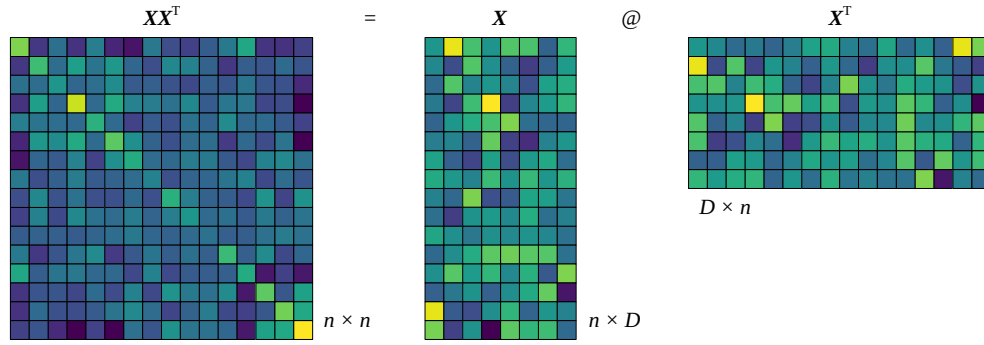
Figure 5.  $U$ 's row vectors are all unit vectors and pairwise orthogonal

⚠ Note that in the full SVD,  $U$  is an orthogonal matrix. In other SVD variants, such as the economy SVD,  $U$  is a semi-orthogonal matrix.

### $XX^T$ 's Eigenvalue Decomposition

As shown in Figure 6,  $XX^T$  is equivalent to the Gram matrix of matrix  $X$ .

Let's review this book's earlier discussion of Gram matrices. A Gram matrix is symmetric, and is either positive semi-definite or positive definite.

Figure 6. Computing the Gram Matrix  $XX^T$ 

From the equation above,  $XX^T$  can be written as

$$XX^T = (USV^T)(USV^T)^T = US \underset{I}{V^T V} S^T U^T = USS^T U^T$$

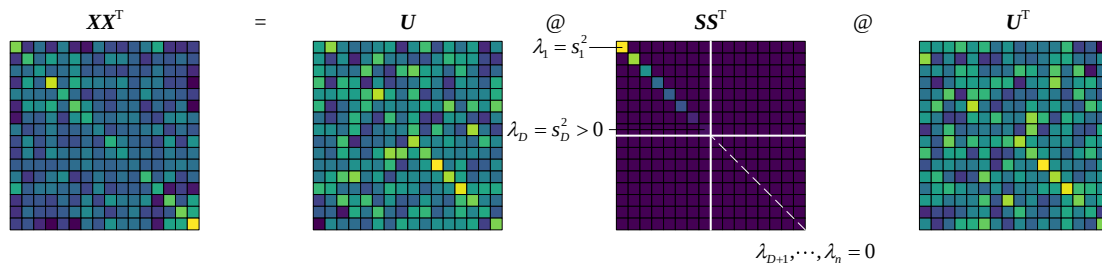
Do you notice that the equation above is exactly an eigenvalue decomposition!

To be precise, since  $XX^T$  is a real symmetric matrix, the equation above is in fact a spectral decomposition, which expands to give

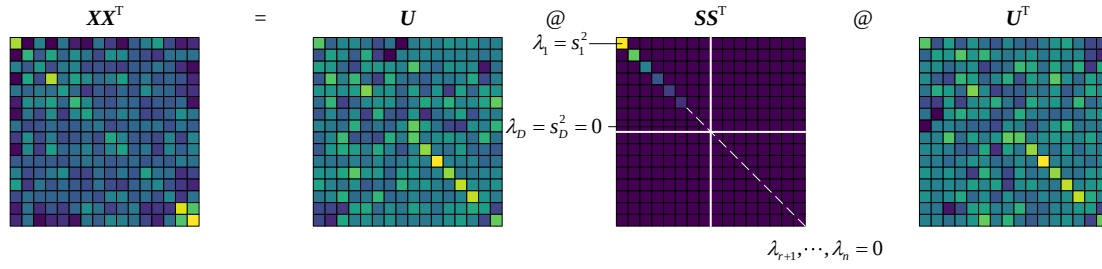
$$XX^T = U \begin{bmatrix} s_1^2 & & & \\ & \ddots & & \\ & & s_D^2 & \\ & & & 0 \\ & & & & \ddots \\ & & & & & 0 \end{bmatrix} U^T$$

You may notice that  $XX^T$ 's eigenvalues include a large number of 0s.

As shown in Figure 7, if  $X$  is full column rank,  $X$ 's singular values are all nonzero, so  $XX^T$  has  $n - D$  eigenvalues equal to 0.

Figure 7.  $XX^T$ 's Eigenvalue Decomposition,  $X$  is Full Column Rank

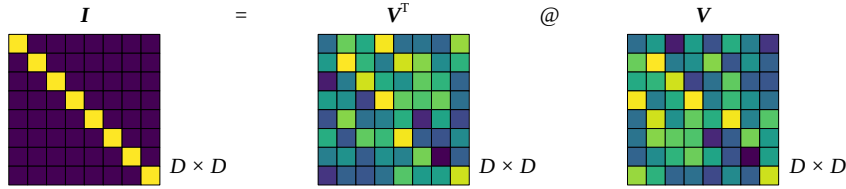
As shown in Figure 8, if  $X$  is not full column rank — say  $\text{rank}(X) = r$  — then  $XX^T$  has  $n - r$  eigenvalues equal to 0.

Figure 8.  $XX^T$ 's Eigenvalue Decomposition,  $X$  is Not Full Column Rank

### $V$ : Right-singular vector matrix

In the full SVD, the right-singular vector matrix  $V$  is an orthogonal matrix, that is, it satisfies

$$V_{D \times D}^T @ V_{D \times D} = V_{D \times D} @ V_{D \times D}^T = I_{D \times D}$$

Figure 9. Right-Singular Vector Matrix  $V$  is an Orthogonal Matrix

Expand using the first perspective of matrix multiplication:  $V^T V$

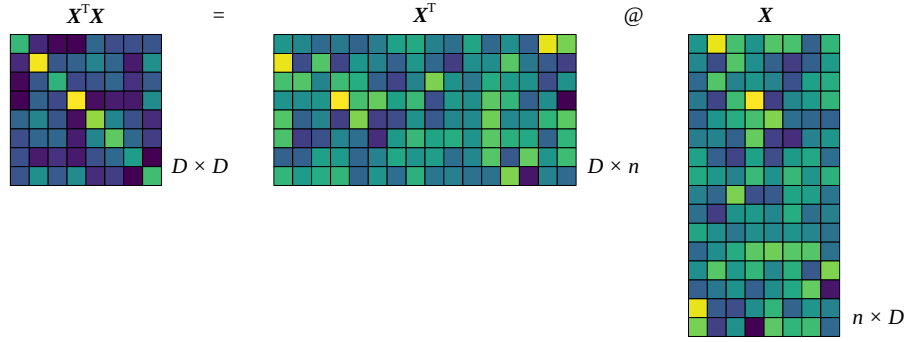
$$V_{D \times D}^T @ V_{D \times D} = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_D^T \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_D \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{v}_D \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \cdots & \mathbf{v}_2^T \mathbf{v}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_D^T \mathbf{v}_1 & \mathbf{v}_D^T \mathbf{v}_2 & \cdots & \mathbf{v}_D^T \mathbf{v}_D \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

In other words, geometrically, the orthogonal matrix  $V$  can be viewed as a rotation in the  $\mathbb{R}^D$  space.

**⚠** Note that in general,  $U$  and  $V$  are rotations in different spaces.

### $X^T X$ 's Eigenvalue Decomposition

As shown in Figure 10,  $X^T X$  is equivalent to the Gram matrix of matrix  $X$ .

Figure 10. Computing the Gram Matrix  $X^T X$ 

From the equation above,  $X^T X$  can be written as

$$X^T X = (USV^T)^T (USV^T) = VS^T \underset{I}{U^T U} SV^T = VS^T SV^T$$

Do you notice that the equation above is also a spectral decomposition!



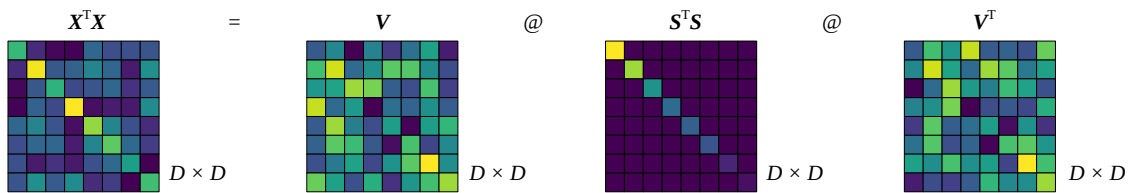
We discussed this spectral decomposition earlier in this book, when introducing Gram matrices.

The equation above can be written as

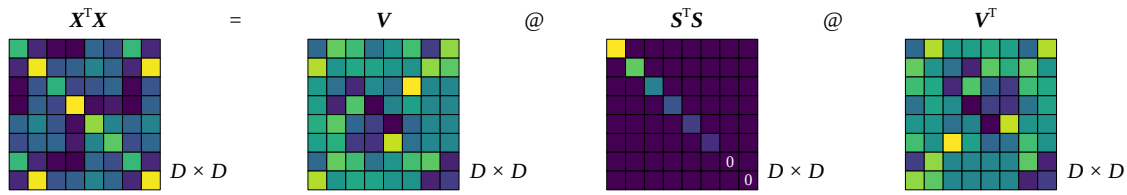
$$X^T X = V \begin{bmatrix} s_1^2 & & & \\ & s_2^2 & & \\ & & \ddots & \\ & & & s_D^2 \end{bmatrix} V^T$$

As shown in Figure 11, if  $X$  is full column rank, then  $X^T X$ 's eigenvalues are all nonzero.

Comparing the two equations above, we find that the two Gram matrices share the same nonzero eigenvalues.

Figure 11.  $X^T X$ 's Eigenvalue Decomposition,  $X$  is Full Column Rank

As shown in Figure 12, if  $X$  is not full column rank — say  $\text{rank}(X) = r$  — then  $X^T X$  has  $D - r$  eigenvalues equal to 0.

Figure 12.  $X^T X$ 's Eigenvalue Decomposition,  $X$  is Not Full Column Rank

### Solving SVD by Hand

Figure 7 and Figure 11 actually give us the idea for solving SVD by hand — spectrally decompose the two Gram matrices!

Next, let's use an example to explain how to solve SVD by hand.

Given the following tall-and-thin matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

To solve for  $V$ , let's first compute the first Gram matrix —  $A^T A$ ,

$$A^T A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}^T \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Further computation gives  $A^T A$ 's eigenvalues and eigenvectors:

$$\left\{ \begin{array}{l} \lambda_1 = 3 \\ \mathbf{v}_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \end{array} \right\} \quad \left\{ \begin{array}{l} \lambda_2 = 1 \\ \mathbf{v}_2 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \end{array} \right\}$$



For solving eigenvalue decomposition by hand, please review Chapter 11, Section 1 of this book.

Then, let's compute the second Gram matrix —  $AA^T$ ,

$$AA^T = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

**⚠** Note the distinction:  $A^T A$  has shape  $2 \times 2$ , while  $AA^T$  has shape  $3 \times 3$ .

Compute  $AA^T$ 's eigenvalues and eigenvectors:

$$\left\{ \begin{array}{l} \lambda_1 = 3 \\ \mathbf{u}_1 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} \end{array} \right\} \left\{ \begin{array}{l} \lambda_2 = 1 \\ \mathbf{u}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ -\frac{\sqrt{2}}{2} \end{bmatrix} \end{array} \right\} \left\{ \begin{array}{l} \lambda_3 = 0 \\ \mathbf{u}_3 = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix} \end{array} \right\}$$

The singular value matrix  $\mathbf{S}$  is as follows:

$$\mathbf{S} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Both equations above give us the eigenvalues  $\lambda_1$  and  $\lambda_2$ .

The singular value matrix  $\mathbf{S}$ 's diagonal elements are the square roots of  $\lambda_1$  and  $\lambda_2$ . This is both an important difference and an important connection between eigenvalue decomposition and SVD.

Therefore,  $\mathbf{A}$ 's full SVD is:

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{\sqrt{3}}{3} \\ \frac{1}{\sqrt{6}} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}^T$$

Solving SVD by hand is not the learning goal of this chapter. What you need to master is the mathematical thinking behind singular value decomposition, and how to understand SVD from different perspectives. In the next section, we'll understand singular value decomposition from a geometric perspective.



LA\_14\_01\_01.ipynb replicates the SVD result using the method above of spectrally decomposing the two Gram matrices. The code also uses `numpy.linalg.svd()` to complete the singular value decomposition computation.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please solve the singular value decomposition of the following matrix by hand, and verify the result using Python.


$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & -1 \\ -1 & 0 \end{bmatrix}$$


$$\mathbf{A} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$


$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$


$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$